

Disproof of TLMC Conjecture 00000001040

A counterexample at the prime $q = 5$

TLMC submission

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Conjecture 1 (00000001040). *The complete-mapping polynomials (polynomials f with $f(x) + x$ a permutation) at $q \equiv 2 \pmod{3}$ form the complete equivalence class of x^3 type (a cubic classification of complete mappings).*

Proposition 1 (Counterexample). *Let $q = 5$, which is prime and satisfies $5 \equiv 2 \pmod{3}$. Then:*

1. $f(x) = 2x$ is a complete mapping of $\mathbb{F}_5$: both $2x$ and $f(x) + x = 3x$ are permutations, with $2x \mapsto (0, 2, 4, 1, 3)$ and $3x \mapsto (0, 3, 1, 4, 2)$;
2. $x^3 + x$ is not a permutation of $\mathbb{F}_5$: it sends $x = 0, 2, 3$ all to 0 (values $(x^3 + x)_{x=0}^4 = (0, 2, 0, 0, 3)$);
3. more strongly, no member of the affine x^3 -family $A(x + B)^3 + C$ with $A \in \mathbb{F}_5^\times$, $B, C \in \mathbb{F}_5$ is a complete mapping;
4. nevertheless exactly 15 linear complete mappings $ax + b$ exist, with slopes $a \in \{1, 2, 3\}$ and $b \in \mathbb{F}_5$, and a brute-force census shows $\mathbb{Z}_5$ has exactly 15 complete mappings in total.

Hence at $q = 5$ the x^3 -type class contains no complete mapping while 15 complete mappings exist: the classification of Conjecture 00000001040 fails. $\square$

Proof. The arithmetic claims are finite computations over $\mathbb{F}_5$; we verify them by hand and mechanically.

(1) $(2x)_{x=0}^4 = (0, 2, 4, 1, 3)$ and $(3x)_{x=0}^4 = (0, 3, 1, 4, 2)$ are both permutations of $\{0, 1, 2, 3, 4\}$, so $f = 2x$ satisfies the complete mapping condition $f + x = 3x$ permutation.

(2) $(x^3 + x)_{x=0}^4 = (0, 2, 10, 30, 68) \equiv (0, 2, 0, 0, 3) \pmod{5}$: the values at $x = 0, 2, 3$ collide at 0, so x^3 is not a complete mapping at $q = 5$.

(3) For $f = A(x + B)^3 + C$ we have $f + x = Ay^3 + y + (C - B)$ with $y = x + B$; since adding a constant and precomposing with the bijection $x \mapsto x + B$ preserve permutation-ness, f is a complete mapping iff $Ay^3 + y$ is a permutation. Direct evaluation for $A \in \mathbb{F}_5^\times$:

- $A = 1$: $(0, 2, 0, 0, 3)$ — 0, 2, 3 collide;
- $A = 2$: $(0, 3, 3, 2, 2)$ — 1, 2 and 3, 4 collide;
- $A = 3$: $(0, 4, 1, 4, 1)$ — 1, 3 and 2, 4 collide;
- $A = 4$: $(0, 0, 4, 1, 0)$ — 0, 1 and 4 repeat 0.

None is a permutation, so the entire affine x^3 -class is free of complete mappings at $q = 5$.

(4) $ax + b$ is a permutation iff $a \neq 0$, and $(a + 1)x + b$ is a permutation iff $a + 1 \neq 0 \pmod{5}$; hence $a \in \{1, 2, 3\}$, $b \in \mathbb{F}_5$: exactly $3 \cdot 5 = 15$ linear complete mappings. A brute-force census over all $5! = 120$ permutations of $\mathbb{F}_5$ (every function $\mathbb{F}_5 \rightarrow \mathbb{F}_5$ is a polynomial of degree < 5) confirms that the total number of complete mappings of $\mathbb{Z}_5$ is exactly 15, i.e. precisely the linear ones.

Since complete mappings exist at $q = 5$ but the x^3 -type equivalence class contains none (and, *a fortiori*, the linear complete mappings are not cubic-type under any degree-preserving equivalence such as affine or EA-equivalence), the classification claimed by the conjecture is false. $\square$

Remark 1 (Boundary). • $q = 5$ is the smallest non-degenerate prime power with $q \equiv 2 \pmod{3}$ ($q = 2$ is degenerate: $\mathbb{F}_2$ admits no complete mapping at all, so the conjecture holds vacuously there).

- The same obstruction persists at every prime $q \equiv 2 \pmod{3}$, $q \geq 5$: the linear map $f = 2x$ has $f + x = 3x$ a permutation (since $3 \nmid q$), and being of degree 1 it cannot lie in the degree-3 x^3 equivalence class; meanwhile x^3 itself is a complete mapping iff $x^3 + x$ is a permutation (e.g. it is one at $q = 11$, where -1 is a quadratic non-residue), which does not rescue a classification that must also include the linear complete mappings.

Mechanical verification

The bundled artifacts reproduce every number above without third-party dependencies:

- `reproduce.py` — independent Python recomputation (exits 0 iff all checks pass; prints the five value-tuples listed in the proof);
- `lean4/` — a Mathlib-free Lean 4 (v4.33.1) development whose main theorem `TLMC1040.disproof_0000000` asserts the conjunction of the numerical facts (1)–(4) together with the explicit 25-case classification of the linear candidates ($a \in \{0, 1, 2, 3, 4\} \times b \in \mathbb{F}_5$), all discharged by `decide`. `lake env lean Check.lean` audits every theorem with `#print axioms`: all report “does not depend on any axioms” (no `sorry`, no `native_decide`).